

YIFAN ZHAO

✉ yifanz16@illinois.edu · 🔗 <https://evzh.net> · in [yifan-zhao-287745162](#) · ☎ (810) 333-7919

RESEARCH INTERESTS

Machine Learning Systems and Compilers; Compiler Design and Optimizations; Performance Engineering; Approximate Computing; Edge Computing.

RESEARCH AND WORK EXPERIENCE

University of Illinois Urbana-Champaign, Urbana, IL 2019 — 2026 (planned)

Graduate Research Assistant, Ph.D. candidate in Computer Science

- Designed and build tensor compilers that optimize DNN inference on GPUs, CPUs, and accelerators.
- Developed the [Neptune](#) tensor compiler: generalizes and automates FlashAttention-style operator fusion; achieves SOTA performance on modern GPUs.
- Developed the [Felix](#) tensor program autotuner: create a differentiable search space for tensor programs, and optimize them with gradient descent.
- Published first-author papers at top ML systems conferences: [Neptune \[PLDI'26\]](#), [Felix \[ASPLOS'24\]](#), [ApproxCaliper \[MLSys'23\]](#).

Nvidia, *Deep Learning Engineer Intern*, Santa Clara, CA May 2024 — Aug. 2024

- Helped develop loop optimizations for an internal DL compiler (polyhedral transformations in MLIR).
- Implemented complex DNN operators (e.g., deformable convolution/attention) in this compiler to probe missing features and performance issues, and deliver usable kernels.
- Wrote an internal paper documenting the improvements.

University of Michigan, *Undergraduate Research Assistant*, Ann Arbor, MI May 2018 — Sep. 2018

- Co-led the Huron project: a compiler analysis and runtime for detecting and repairing false sharing (performance bug) in parallel CPU programs.
- Published a co-first-author paper on Huron at [PLDI'19](#).

EDUCATION

University of Illinois Urbana-Champaign, Urbana, IL 2019 — Now

Ph.D. candidate in Computer Science; **Advisors**: Sasa Misailovic, Vikram Adve

Selected courses: Advanced Compiler Design, Compilers for Edge Computing, ML for Software Engineering

University of Michigan, Ann Arbor, MI 2017 — 2019

Bachelor of Science and Engineering in Computer Science

Awards: James B. Angell Scholar, University Honors, Dean's Honor List

Shanghai Jiaotong University, Shanghai, China 2015 — 2019

Bachelor of Science in Electrical and Computer Engineering

Shanghai Jiaotong — Univ. of Michigan Dual Degree Program

TECHNICAL SKILLS

- Areas of expertise: compilers (deep learning and general-purpose), programming languages, system design, performance engineering
- Compilers: LLVM; TVM, MLIR; Triton, JAX Pallas, TileLang
- Deep learning frameworks: PyTorch (incl. TorchInductor), JAX; CUTLASS
- Programming languages: Python, C/C++, CUDA, Rust (working languages); also familiar with Haskell, C#, JavaScript.

PUBLICATIONS

- [PLDI'26] **Neptune: Advanced ML Operator Fusion for Locality and Parallelism on GPUs.**
Yifan Zhao, Egan Johnson, Prasanth Chatarasi, Vikram Adve, Sasa Misailovic
- [Under review] **Nautilus: Auto-Scheduling Tensor Compiler for Advanced Fusion in Asynchronous and Pipelined GPU Kernels.**
Yifan Zhao, Yuchen Yang, Matei Budiu, Sasa Misailovic
- [ISSTA'26] **ARQ: A Mixed-Precision Quantization Framework for Accurate and Certifiably Robust DNNs.**
Yuchen Yang, Yifan Zhao, Shubham Ugare, Gagandeep Singh, Sasa Misailovic
- [ASPLOS'24] **Felix: Optimizing Tensor Programs with Gradient Descent.**
Yifan Zhao, Hashim Sharif, Vikram Adve, Sasa Misailovic
- [PACMI'24] **Programming Systems for Improving the Performance of Deep Learning-Intensive Applications.**
Yifan Zhao, Vikram Adve, Sasa Misailovic
- [IEEE IoT'24] **Mobiprox: Supporting Dynamic Approximate Computing on Mobiles**
Matevz Fajjancic, Octavian Machidon, Hashim Sharif, Yifan Zhao, Sasa Misailovic, Veljko Pejovic
- [MLSys'23] **ApproxCaliper: Exploiting Application-level Error Resiliency for Optimizing Neural Networks.**
Yifan Zhao*, Hashim Sharif*, Peter Pao-Huang, Vatsin Ninad Shah, Arun Narenthiran, Mateus Valverde Gasparino, Nathan Zhao, Abdulrahman Mahmoud, Sarita Adve, Girish Chowdhary, Sasa Misailovic, Vikram Adve
- [VR'23] **Power, Performance, and Image Quality Tradeoffs in Foveated Rendering.**
Rahul Singh, Muhammad Huzaifa, Jeffrey Liu, Anjul Patney, Hashim Sharif, Yifan Zhao, Sarita Adve
- [IEEE Micro'22] **HPVM: Hardware-Agnostic Programming for Heterogeneous Parallel Systems**
Adel Ejje, Aaron Councilman, Akash Kothari, Maria Kotsifakou, Leon Medvinsky, Abdul Rafae Noor, Hashim Sharif, Yifan Zhao, Sarita Adve, Sasa Misailovic, Vikram Adve
- [PPoPP'21] **ApproxTuner: A Compiler and Runtime System for Adaptive Approximations.**
Hashim Sharif, Yifan Zhao, Maria Kotsifakou, Akash Kothari, Ben Schreiber, Elizabeth Wang, Yasmin Sarita, Nathan Zhao, Keyur Joshi, Vikram Adve, Sasa Misailovic, Sarita Adve
- [PLDI'19] **Huron: Hybrid False Sharing Detection and Repair.**
Tanvir Ahmed Khan*, Yifan Zhao*, Gilles Pokam, Barzan Mozafari, Baris Kasikci

TEACHING AND MENTORING

Teaching Assistant

- Fall 2023 UIUC CS 426 Compiler Construction (Sasa Misailovic)
Fall 2018 UMich CS 482 Operating Systems (Baris Kasikci)

Guest Lectures

- Fall 2023 UIUC CS 426 Compiler Construction (2 tutorial lectures)
Spring 2022 UIUC CS 598 Approximate and Probabilistic Computing
Fall 2021 UIUC CS 526 Advanced Compiler Construction

Research Mentoring

- 02/26 – present UIUC undergraduate student, Aryan Agarwal and Sophia Baltz
Collaborating to extend Neptune to new operators and improve scheduling.
05/25 – 12/25 UIUC undergraduate student, Minsoo Kim
Collaborated to extend Neptune to new hardware and operator setups.

- 09/24 – 01/25 UIUC Undergraduate (IBM IIDAI program), Krish Patel and Vishesh Prasad
Collaborated to improve the latency-prediction cost model used in Felix.
- 08/22 – 01/23 UIUC Ph.D. student, Vimarsh Sathia
Collaborated on an extension of the ApproxCaliper project.
- 12/20 – 12/22 UIUC undergraduate student, Peter Pao-Huang
Collaborated on the ApproxCaliper project.
- 12/21 – 05/22 UIUC undergraduate student, Yi Zhou
Collaborated on an accuracy-aware tensor compiler project.

TALKS

Neptune: Advanced ML Operator Fusion for Locality and Parallelism on GPUs

- PLDI'26 conference talk, 06/19/2026, Boulder
- Talk to IBM Compiler Team, 04/20/2026, Virtual
- Compiler Seminar talk @ Illinois, 10/02/2025, University of Illinois

Felix: Optimizing Tensor Programs with Gradient Descent

- ASPLOS'24 conference talk, 04/30/2024, San Diego
- Talk to IBM Compiler Team, 12/08/2023, Virtual

ApproxCaliper: Exploiting Application-level Error Resiliency for Optimizing Neural Networks

- MLSys'23 conference talk, 06/06/2023, Miami
- UIUC Compiler Seminar, 02/07/2022, University of Illinois
- Illinois Autonomous Farm Workshop, 07/07/2021, University of Illinois

ApproxTuner: A Compiler and Runtime System for Adaptive Approximations

- Invited talk for a CS 598 (approx. computing) lecture, 03/10/2022, University of Illinois
- UIUC Visit Day Poster Session, 03/06/2022, University of Illinois
- Talk to Qualcomm Compiler Team, 12/13/2020, Virtual
- Talk to Amazon Compiler Team, 07/24/2020, Virtual

CONTRIBUTIONS TO RESEARCH GRANTS

Significantly contributed to writing of the proposals that led to the following research grants being awarded:

Fellini: Differentiable ML Compiler for Full-Graph Optimization for LLM Models

- Amazon Research Award; PIs: Sasa Misailovic, Vikram S. Adve; Apr. 2025.
- \$120k in cash funding and credits.

Felix: Gradient-Based Compiler Optimization of Tensor Programs

- IBM-Illinois Discovery Accelerator Institute (IIDAI); PIs: Sasa Misailovic, Vikram S. Adve; May 2023.
- Amount: \$500k.

REFERENCES

Sasa Misailovic

Associate Professor
Siebel School of Computer Science, UIUC
misailo@illinois.edu

Tianyin Xu

Associate Professor
Siebel School of Computer Science, UIUC
tyxu@illinois.edu

Vikram Adve

Donald B. Gillies Professor
Siebel School of Computer Science, UIUC
vadve@illinois.edu

Charith Mendis

Assistant Professor
Siebel School of Computer Science, UIUC
charithm@illinois.edu